

Tushar Nayak

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Computer Vision and Deep Learning Researcher/Engineer focused on multimodal perception, 2D–3D registration and reconstruction, geometric & physics-informed deep learning. Building real-time medical systems that recover structure from sparse visual observations.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science - Biomedical Engineering (Research) – [GPA 3.8/4.0]

May 2026

- COURSES: Computer Vision, Learning for 3D Vision, Computational Modeling & Analysis, Medical Image Analysis.
- TA: Computer Vision, Applied Deep Learning, ML in Biomedical Research, Fundamentals of Computational Bioengineering.
- AWARDS AND LEADERSHIP: CMU-BME Department Head's Fellowship, CMU-BME Department Ambassador.

Manipal Institute of Technology

Manipal, India

Bachelor of Technology - Biomedical Engineering, Minor in Data Science

May 2023

- AWARDS: Best Paper* - 2nd International Conference on Artificial Intelligence, Computational Electronics & Communication
- LEADERSHIP: IEEE-EMBS Student Chapter Chair & Research Head; IEEE-RAS Student Chapter Exec. Board & Webmaster.

SKILLS

Deep Learning: GNNs, Neural ODEs, PINNs, Attention Mechanisms & Transformers, U-Nets, VLMs, Occupancy Networks.

3D Vision: Reconstruction, Neural Rendering, Deformable Registration, Implicit Representations, Differential Rendering.

Computer Vision: Detection, Segmentation, Multi-view Geometry, Camera Geometry, Optical Flow, Structure from Motion.

Frameworks: PyTorch, PyTorch3D, PyGeometric, TorchDiffEq, TensorFlow, Keras, CUDA, OpenCV, ITK, MONAI, Git, Docker.

Languages: Python, MATLAB, C++, R, Julia, (Linux), Bash, $\LaTeX$.

RESEARCH

Computational Engineering and Robotics Lab, Carnegie Mellon University

Pittsburgh, PA

Graduate Researcher, Real-time 3D Vascular Deformation Estimation 

Aug 2024 – present

- Architected FLUOROGRID, a **multi-modal quad-encoder graph neural network** fusing live 2D fluoroscopy feeds with 3D pre-operative volumes to **map 3D deformations from 2D input in real time** to reduce continuous operative radiation.
- Engineered a geometrically regularized loss function leveraging discrete differential geometry to enforce structural limits, achieving **sub-millimeter precision** and real-time inference speeds (**>120fps on CUDA, >30fps on CPU**).
- Built a physics-grounded synthetic data pipeline **simulating viscoelastic tool-tissue interactions and fluoroscopy ray-tracing**, generating an expansive dataset of **370,000+ 3D–2D training data pairs**.

The $\forall$ Lab and Image Science Lab, Carnegie Mellon University

Pittsburgh, PA

Research Assistant, Longitudinal Glioblastoma Evolution Visual Forecasting 

Feb 2025 – Aug 2025

- Developed a **Neural ODE-based forecasting pipeline** to visually predict longitudinal glioblastoma growth using multimodal MRI.
- Built an **Attention U-Net & temporal embedding model** to simulate the evolution of latent tumor in continuous time.
- Presented findings at the **CMU BME Forum 2025** , currently modeling tumor evolution as an explicit dynamics problem.

Biomedical Informatics Computing Lab, Manipal Institute of Technology

Manipal, India

Undergraduate Researcher, Multi-Stage Multi-Modal Oral Cancer Detection

Jan 2022 – May 2023

- Designed an ensemble model combining **3 transfer-learning models with self-attention** mechanisms for oral cancer detection.
- Achieved **95% accuracy** across lesion photos, OCT, and histological modalities, with class activation mapping for explainability.
- Published & presented research findings on this pipeline, alongside diagnostic models for Lung Cancer, Mpox, & Dengue* .

PROJECTS

Sparse 2D Echocardiograph to 3D Cardiac Volume Reconstruction



(CMU)

Aug 2025 - present

- Developed **coordinate-based neural implicit representation** model for cardiac volume reconstruction with surface priors from only three sparse 2D echocardiography slices achieving **3D Dice 0.864 and 3D IoU 0.765** on the MITEA-CAP dataset.
- Implemented **differentiable multi-view projection with learnable pose optimization** for joint slice alignment & geometry recovery, used Fourier positional encoding to capture high-frequency boundaries under sparse supervision.
- Exploring a **stabilized Gaussian occupancy field** pipeline, clubbing differentiable rasterization and pose optimization.

Deformable 3D Reconstruction of Live Endoscopic Surgical Scenes



(CMU)

Apr 2026 - present

- Prototyping a **deformable 3D Gaussian Splatting pipeline with VLM semantic embeddings** for endoscopic scenes.
- Preliminary results show a +8.5dB PSNR, 75% depth error reduction, and initial zero-shot text querying on ENDOSCAPHES dataset.

Dual View Mammogram-based Breast Cancer Prediction



(CMU)

Jan 2026 - May 2026

- Developed a **spatial alignment pipeline** to register MLO mammograms to CC view using breast-mask **centroid matching and affine translation**, for better dual-view feature correspondence by **reducing cross-view centroid distance by 99.91%**.
- Built a **cross-view attention classifier** that fused aligned mammograms for binary prediction at **80%** on CBIS-DDSM dataset.

Neural PDE-based Anisotropic Diffusion for MRI Denoising



(CMU)

Feb 2026- Mar 2026

- Unrolled Perona-Malik PDE with **learned conduction weights**, achieving **+23% PSNR** and **47% lower Edge MSE** on BR35H.

Perception and Visual Learning



(Education Team, Open Horizon Robotics)

Mar 2026 - present

- Authoring a **comprehensive open-source computer vision course** covering Classical 2D Vision to Localization & Mapping..